

Deploy post-quantum cryptography hardware for satellite communications to protect space assets from future quantum threats, targeting defense and satellite operators.

Aerospace and defense industry × Post-quantum cryptography migration × Post-quantum cryptography · OS147 v1 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
76 is the world moving	45 can we play, can we win	MEDIUM 1 named in the evidence	€75.6m modelled evidence · per year	NEXT regulation adopted or propose...	L0 14 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

Evidence gap (SC-13). Orange has 0.0 published reference(s) in Aerospace and defense industry, below the threshold of 5. A customer conversation here has no proof point behind it yet.

The opportunity

Deploy post-quantum cryptography hardware for satellite communications to protect space assets from future quantum threats, targeting defense and satellite operators. This opportunity exists now because large-scale quantum computers threaten current public-key cryptography, and adversaries can already record encrypted traffic for future decryption, while the space sector is expanding rapidly with new satellite constellations.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	€91.7m €16.5m – €441m	€75.6m €13.6m – €364m	€1.1m €204k – €5.5m	modelled
Observed: tendered contract value, annualised	€45.7m	€45.7m	€686k	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Aerospace and defense industry (EU27_2020)	3,354 enterprises	observed	Eurostat · sbs_sc_oww	2024
Enterprises adopting Post-quantum cryptography	49.0 % of enterprises (10+ employees)	proxy	Eurostat · isoc_cisce_ran2 · E_SECMDENC	2024
Annual value of one engagement	€240k per enterprise per year	assumption	config/sizing.yaml · fallback_bands_eur	size-2026-08-a
Size-weighted buyer base	780 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	1.5 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Adoption rate is a declared proxy. ICT security measures by NACE Rev. 2 activity series E_SECMDENC is used as a proxy for Post-quantum cryptography: Encryption users are the migration population. Contract value is a configured assumption, not observed. Only 0 matching tender notices, below the 6 needed for a robust median, so the configured band for this business domain is used. It is an assumption owned by innovation-radar-curator, not evidence. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders. 1 of 1 declared geographies are outside the European reference data (US); the estimate covers the European footprint only.

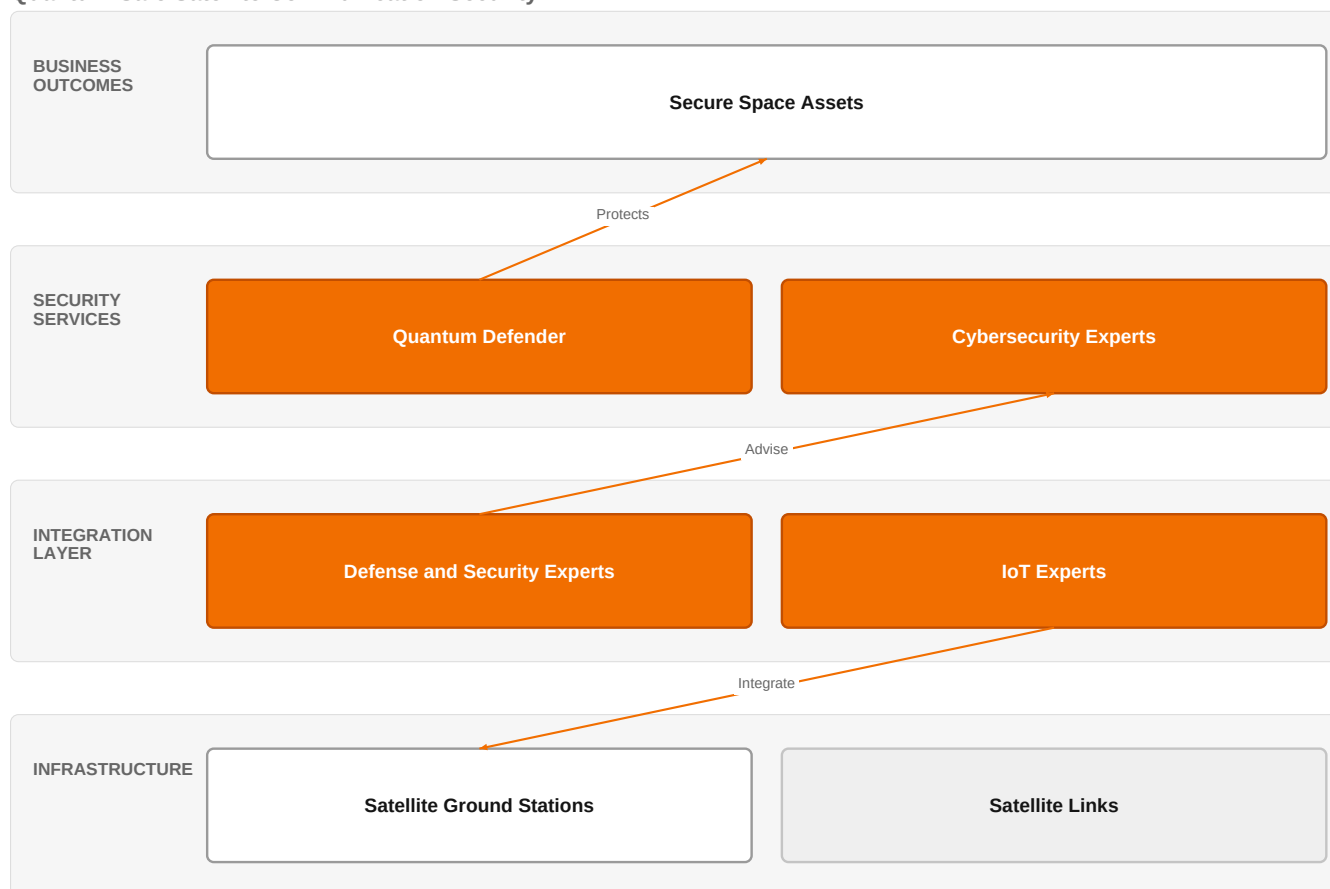
Observed: tendered contract value, annualised

Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€45.7m per year	observed	TED (Tenders Electronic Daily) · ted	2026-07-07..2026-07-08
Assumed obtainable share of the serviceable market	1.5 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 3 matching notices, matched on vertical via the CPV crosswalk — §4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a market size. 1 of 1 declared geographies are outside the European reference data (US); the estimate covers the European footprint only.

The solution, in one picture

Quantum-Safe Satellite Communication Security



This diagram shows how Orange's quantum-safe security services protect satellite communications, integrating with existing ground infrastructure.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

Who buys, and why now

The buyer is typically the Chief Information Security Officer (CISO) or Chief Technology Officer (CTO) at a defense agency or satellite operator. The pain is operational: current cryptographic assets on satellites and ground stations are vulnerable to harvest-now-decrypt-later attacks, and embedded systems on defense platforms are often overlooked in enterprise PQC migration efforts. The trigger is often a regulatory mandate or a security audit that identifies cryptographic assets at risk, or the announcement of new satellite constellations that require quantum-safe security from the start.

Sources: SIG-13AAE2CD702E MDPI AG 2026-01-19; SIG-40B64E273856 International Journal 2026-07-30

Why it is hot now — every claim bound to a dated source

- Northrop Grumman and Aeronix announced a fivefold boost in space encryption hardware performance, indicating a shift toward more capable encryption solutions for space applications. [SIG-7ECDD0A30D62]
- SpaceX seeks FCC approval for 100,000 Gen3 satellites, expanding the space infrastructure that will require robust security. [SIG-6F7D1D98901A]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would deliver a quantum-safe satellite communications security assessment and migration roadmap, leveraging its Cybersecurity experts and Defense and security experts to inventory cryptographic assets, prioritize risks, and design a hybrid PQC migration plan. The engagement would include deploying Quantum Defender / quantum-safe offers to protect satellite links, with Orange Group researchers providing technical guidance. The order would be: assess, design, pilot, and deploy.

Why Orange can win

Orange can win by combining its cybersecurity and defense expertise with its global network and trusted relationships in the aerospace and defense sector. The Quantum Defender / quantum-safe offers provide a concrete solution, and Orange's research capabilities add credibility. However, the assets are thin in terms of specific space-hardware experience, so Orange must partner or leverage its broader ecosystem to compete.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Quantum Defender / quantum-safe offers	offer	L0	offer addresses the use case AND provides the technology	unconfirmed
Cybersecurity experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
IoT experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Digital experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Defense and security experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed

Competitive landscape

MEDIUM competitive intensity (score 6.2; 1 of 8 listed players appear in this topic's own evidence)

A real field, including at least one player the customer will already know. Expect to be compared.

Competitors include SpaceX Starlink, which is expanding its satellite constellation and could bundle security; Atos/Eviden, IBM, Sopra Steria as global systems integrators with PQC offerings; and security specialists like Airbus Protect, Entrust, ID Quantique, and Thales, all active in aerospace and defense. Thales and Airbus Protect have deep domain expertise, while ID Quantique specializes in quantum-safe solutions. The customer will compare Orange's end-to-end integration and trust position against these specialists.

Sources: SIG-6F7D1D98901A techrepublic.com 2026-07-13

Competitor	Category	Position here	Basis	Evidence
SpaceX Starlink	Satellite operator	competes in Connectivity Solutions	evidenced	SIG-6F7D1D98901A
Atos / Eviden	Global systems integrator	sells Post-quantum cryptography; active in Aerospace and defense industry; competes in Cybersecurity	structural	—
IBM	Global systems integrator	sells Post-quantum cryptography; competes in Cybersecurity	structural	—
Airbus Protect	Security specialist	sells Post-quantum cryptography; active in Aerospace and defense industry; competes in Cybersecurity	structural	—
Entrust	Security specialist	sells Post-quantum cryptography; competes in Cybersecurity	structural	—
ID Quantique	Security specialist	sells Post-quantum cryptography; competes in Cybersecurity	structural	—
Thales	Security specialist	sells Post-quantum cryptography; active in Aerospace and defense industry; competes in Cybersecurity	structural	—
Sopra Steria	Global systems integrator	active in Aerospace and defense industry; competes in Cybersecurity	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register

version competitors-2026-08-a.

Qualifying questions for the first meeting

- 1 What cryptographic algorithms are currently used in your satellite communication links, and have you inventoried them?
- 2 Have you experienced any 'harvest now, decrypt later' concerns or been advised by regulators to start PQC migration?
- 3 What is the expected lifespan of your current satellites, and will they need to be quantum-safe before decommission?
- 4 Do you have a roadmap for upgrading ground station hardware to support PQC, or is that still undefined?
- 5 How are you currently addressing security for embedded systems on your platforms, and is PQC part of that plan?

Objections you will meet

They will say	You can answer
Quantum computers are still years away; why invest now?	It's true that large-scale quantum computers are not yet operational, but adversaries can already record encrypted traffic to decrypt later, and the migration itself takes years. Starting now ensures you're not caught unprepared.
We already have a security partner for PQC.	Many security partners offer PQC, but Orange brings a unique combination of cybersecurity expertise, defense domain knowledge, and a trusted position in the aerospace and defense sector, plus our Quantum Defender offers.
Space hardware is highly specialized; can Orange really deliver?	We acknowledge that space-grade hardware is specialized, but we can leverage our research and partnerships to provide a solution that meets the rigorous requirements, and we can start with an assessment to prove our capability.

Risks and unknowns

The main risk is that PQC standards are still evolving, and hardware for space is specialized; Orange may lack proven space-grade hardware. Also, the timeline for CRQCs is uncertain, so customers may delay investment. The migration of embedded systems is complex and could stall without clear regulatory deadlines.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Bid brief for a defense or satellite operator RFP, differentiating Quantum Defender / quantum-safe offers by highlighting Orange's cybersecurity, IoT, digital, and defense expertise, and referencing the regulatory push for post-quantum cryptography.
Sales	In a meeting with a defense or satellite operator's CISO, open with: 'Given the recent performance gains in space encryption hardware and the scale of new satellite constellations, how are you planning for quantum-safe security in your space assets?'
Strategist / Innovator	Deep-dive brief on how to position Quantum Defender / quantum-safe offers against the fivefold performance boost announced by Northrop Grumman and Aeronix, focusing on defense and satellite operators as the target segment.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-40B64E273856	2026-07-30	International Journal of ...	1	Post-Quantum Cryptography Migration for Enterprise Security
SIG-B8561992126B	2026-07-30	openalex.org	1	Post-Quantum Cryptography Migration for Enterprise Security
SIG-BFFA5CE48BF5	2026-02-27	Shanlax International Jou...	1	Migration Strategies from Traditional Cryptography to Post-Quantum Cryptography in Cloud Systems
SIG-A6F5B58EFCE3	2026-02-12	openalex.org	1	Post-Quantum Cryptography in Networking Protocols: Challenges, Solutions, and Future Directions
SIG-13AAE2CD702E	2026-01-19	MDPI AG	1	Cryptographic Asset Discovery and Inventory for Embedded Systems: A Framework for Post-Quantum ...
SIG-53F3508BD517	2026-01-09	openalex.org	1	Post-Quantum Public Key Infrastructures: Hybrid Certificates, Cryptographic Combiners, and Migr...
SIG-CE5A95F7AF72	2026-01-06	openalex.org	1	PQC Planner: Post-Quantum Cryptographic Migration Planner
SIG-353070727CBB	2026-01-01	openalex.org	1	Institutional Approaches to Post-Quantum Cryptography: A Comparative Analysis of Migration Fram...
SIG-D45704585F54	2026-01-01	Elsevier BV	1	Towards a Quantum Risk Index: a risk-based framework for prioritizing Post-Quantum Cryptography...
SIG-7C4F80CBCF14	2025-09-19	openalex.org	1	The Strategic Imperative of Quantum Readiness: A Comprehensive Review of Post-Quantum Cryptogra...
SIG-2153E369E0CC	2025-09-15	Journal of Electrical and...	1	Securing Digital Communications: Quantum and Post-Quantum Cryptography
SIG-43901F3A86B3	2025-04-09	Wiley	1	Post-Quantum Cryptography: Evaluating and Standardizing Quantum-Resistant Algorithms
SIG-7ECDD0A30D62	2026-08-13	techtimes.com	2	Space Encryption Hardware Gets Fivefold Boost : Northrop Grumman and Aeronix
SIG-6F7D1D98901A	2026-07-13	techrepublic.com	2	SpaceX Seeks FCC Approval for 100,000 Gen3 Satellites - techrepublic.com

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: what_is_changing (named entities that were never supplied: Northrop Grumman, Aeronix)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:14+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:17+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-19T04:57:25+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.